Structural Hardness as Covariance Geometry in Random 3-SAT

A MAAT Scaling Benchmark for Runtime Ridges and Defect Conditioning

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2026

Abstract

Random 3-SAT exhibits a well-known hardness peak near the satisfiability transition. This note tests whether such hardness can be described in terms of structural diagnostics rather than satisfiability alone. Using a MAAT-based representation of random 3-SAT instances, we compute five primitive structural fields: harmony, balance, activity, connectivity, and respect. These are converted into defect variables whose covariance geometry is analysed across ensembles of formulas.

Two experiments are reported. First, a local instance-level classifier tests whether k-nearest-neighbour defect covariance improves prediction of top-20% runtime-hard instances. The result is weak: local covariance features carry information, but do not yet produce a strong standalone classifier. Second, a scaling benchmark over $n = 60, 90, 120, 180$ variables and clause densities $\alpha \in \{3.8, 4.1, 4.26, 4.4, 4.8\}$ confirms a clear runtime ridge near $\alpha \approx 4.26$–4.4 that grows strongly with problem size.

The main conclusion is cautious. Defect covariance geometry is a useful ensemble-level diagnostic, but raw condition number $\kappa(C)$ is not a universal scalar hardness law. In the present benchmark, connectivity V is the most stable within-size ridge marker, while covariance volume and effective rank provide secondary structural information.

1 Introduction

The satisfiability problem is a canonical setting for studying computational hardness. Random 3-SAT in particular exhibits a sharp transition in satisfiability probability and a corresponding region of elevated solver runtime near the critical clause density [1, 2, 3, 4]. Classically, this transition is described in terms of the clause-to-variable ratio

$$\alpha = \frac{m}{n}.$$

The MAAT structural-selection programme asks a complementary question: can hardness be described not only by α or satisfiability, but by the geometry of structural defects? The guiding hypothesis is:

> *SAT runtime hardness may be organised by the covariance geometry of primitive structural defects.*

This note tests that hypothesis empirically. The result is deliberately framed as a benchmark, not as a proof of complexity-theoretic separation. In particular, the experiments do not prove $P \neq NP$, do not establish a universal hardness law, and do not replace standard SAT theory. They provide a reproducible structural diagnostic experiment.

2 MAAT Representation of Random 3-SAT

Each random 3-SAT instance is represented by five structural fields

$$(H, B, S, V, R) \in [0, 1]^5.$$

In this benchmark, the operational definitions are:

- H: clause-count coherence relative to the number of variables;
- B: literal balance between positive and negative occurrences;
- S: structural diversity of generated clauses;
- V: sparse variable-connectivity induced by co-occurrence in clauses;
- R: variable-usage fairness, used as a robustness proxy.

The derived legacy projection is

$$\widehat{C} = H\,B\,S\,V\,\frac{1}{1+R} + \epsilon. \tag{1}$$

This projection is not assumed to be a universal hardness law. It is included as a comparison feature.

Primitive defects are defined by

$$d_a = 1 - a, \qquad a \in \{H, B, S, V, R\}. \tag{2}$$

For an ensemble $\mathcal{E}$ of formulas, the defect covariance matrix is

$$C_{ab} = \mathrm{Cov}_{\mathcal{E}}(d_a, d_b). \tag{3}$$

We compute the following covariance observables:

$$\kappa(C) = \frac{\lambda_{\max}(C)}{\lambda_{\min}(C) + \varepsilon}, \tag{4}$$

$$\log\det(C) = \sum_i \log_{10}(\lambda_i + \varepsilon), \tag{5}$$

$$r_{\mathrm{eff}}(C) = \exp\left[-\sum_i p_i \log p_i\right], \qquad p_i = \frac{\lambda_i + \varepsilon}{\sum_j (\lambda_j + \varepsilon)}. \tag{6}$$

Here $\kappa(C)$ measures conditioning, $\log\det(C)$ measures covariance volume, and r_{eff} measures effective spectral rank.

3 Experimental Design

The experiments use random 3-SAT formulas generated with n variables and $m = \lfloor \alpha n \rceil$ clauses. Each formula is solved with MiniSat22 through the PySAT interface [5]. The runtime observable is

$$\log_{10}(t + \varepsilon),$$

where t is the wall-clock solve time.

3.1 Local Classifier Benchmark

The first experiment uses the existing 2000-instance SAT atlas with $n = 120$ and $\alpha \in [3.0, 5.0]$. An instance is labelled as runtime-hard if its runtime lies in the top 20% within its α group. This avoids confusing logical satisfiability with computational hardness.

Local covariance features are computed from the $k = 40$ nearest neighbours of each instance in the space

$$(\alpha, d_H, d_B, d_S, d_V, d_R).$$

Four models are compared:

Model	Features
alpha only	α
alpha + $\widehat{C}$	$\alpha, \widehat{C}$
alpha + fields	α, H, B, S, V, R
full local covariance	fields plus local covariance observables

3.2 Scaling Benchmark

The second experiment tests whether ensemble-level covariance geometry scales with hardness across problem sizes. The grid is:

$$n \in \{60, 90, 120, 180\}, \qquad \alpha \in \{3.8, 4.1, 4.26, 4.4, 4.8\},$$

with 60 random instances per (n, α) cell, for a total of 1200 formulas.

4 Results

4.1 Runtime Ridge

The scaling benchmark reproduces the expected runtime ridge near the 3-SAT transition. The hardest median-runtime cell is:

n	α	SAT probability	median runtime	median $\log_{10}$ runtime
180	4.26	0.5333	0.018261	-1.7386

This confirms that the benchmark is probing the relevant hard region.

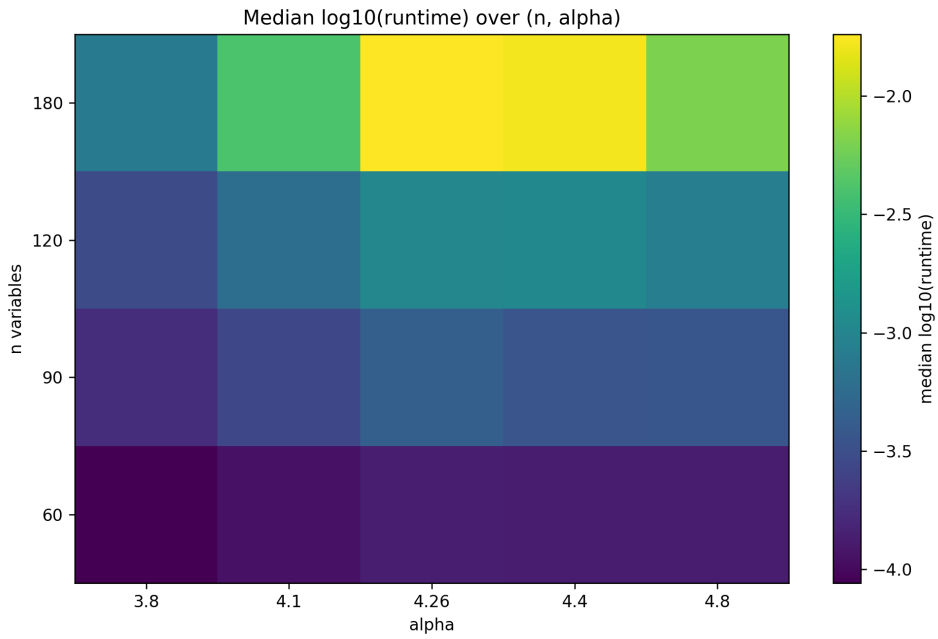

Figure 1: Median $\log_{10}$ runtime across problem size n and clause density α. Runtime increases strongly with n and peaks near the expected transition region $\alpha \approx 4.26$–4.4.

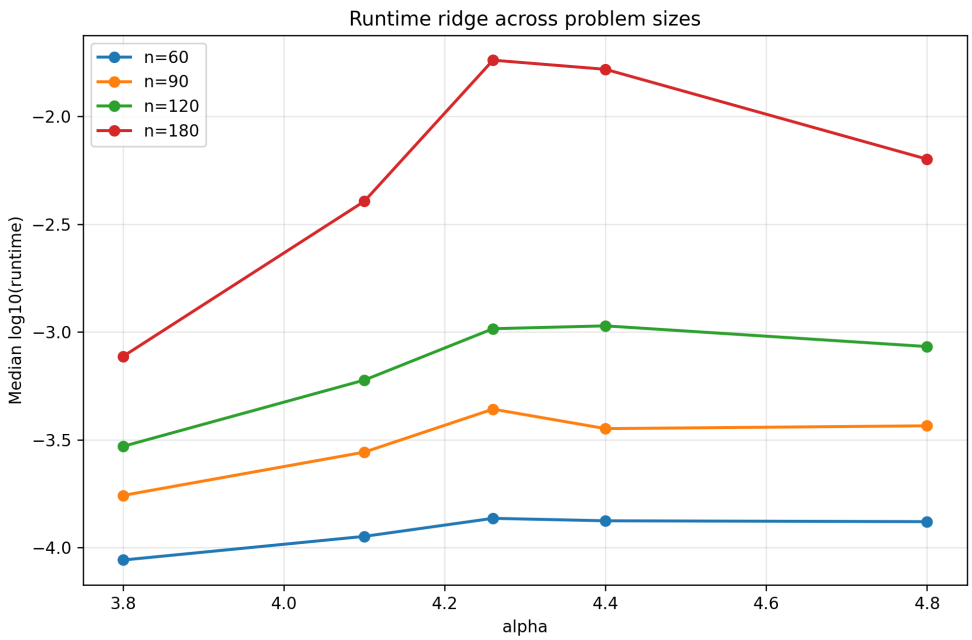

Figure 2: Runtime ridge across problem sizes. The ridge becomes pronounced for $n = 180$, with a maximum near $\alpha = 4.26$.

4.2 Covariance Conditioning

The naive hypothesis

$$\text{hardness} \sim \kappa(C)$$

is not supported as a universal scalar law in the present scaling run. Across all (n, α) cells, raw covariance conditioning correlates negatively with median runtime:

Quantity	Pearson r	Spearman r_s
covariance effective rank	-0.7193	-0.7444
covariance log determinant	-0.6674	-0.8316
covariance log kappa	-0.8288	-0.8722
connectivity V	-0.8071	-0.8286

This negative cross-size correlation does not mean that covariance geometry is irrelevant. Rather, it shows that the raw covariance scale itself changes with problem size. Therefore $\kappa(C)$ cannot yet be interpreted as a size-independent hardness scalar.

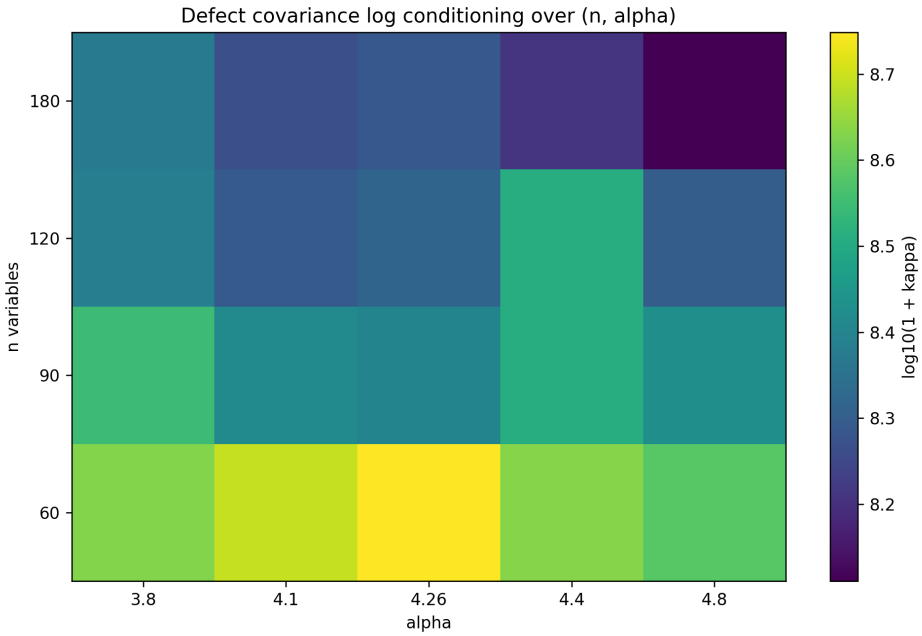

Figure 3: Defect covariance conditioning over (n, α). The raw condition number does not align monotonically with the runtime ridge across problem sizes, indicating that additional size-normalisation is needed.

4.3 Within-Size Structure

Within fixed n, connectivity V is the most stable ridge marker in this moderate sample run:

n	Spearman(V, runtime)	Spearman($\log \kappa$, runtime)	Spearman(r_{eff}, runtime)
60	0.6000	0.5000	-0.3000
90	0.7000	-0.7000	0.5000
120	0.7000	0.4000	-0.3000
180	0.6000	-0.3000	0.5000

Thus the present evidence supports a more careful statement:

> *MAAT connectivity tracks the hardness ridge within fixed problem size, while covariance geometry describes ensemble structure but is not yet a universal scalar hardness law.*

4.4 Local Classifier Benchmark

The local covariance classifier gives a useful negative result. Adding local covariance features improves balanced accuracy only weakly:

Model	ROC-AUC	Accuracy	Balanced accuracy	F1
α only	0.4313	0.3450	0.4500	0.2762
$\alpha + \widehat{C}$	0.4754	0.6183	0.4708	0.1908
α+ fields	0.4665	0.6917	0.4792	0.1395
fields + local covariance	0.4737	0.7517	0.5042	0.1287

The local features are informative in the feature-importance ranking, but do not yet provide a strong instance-level classifier.

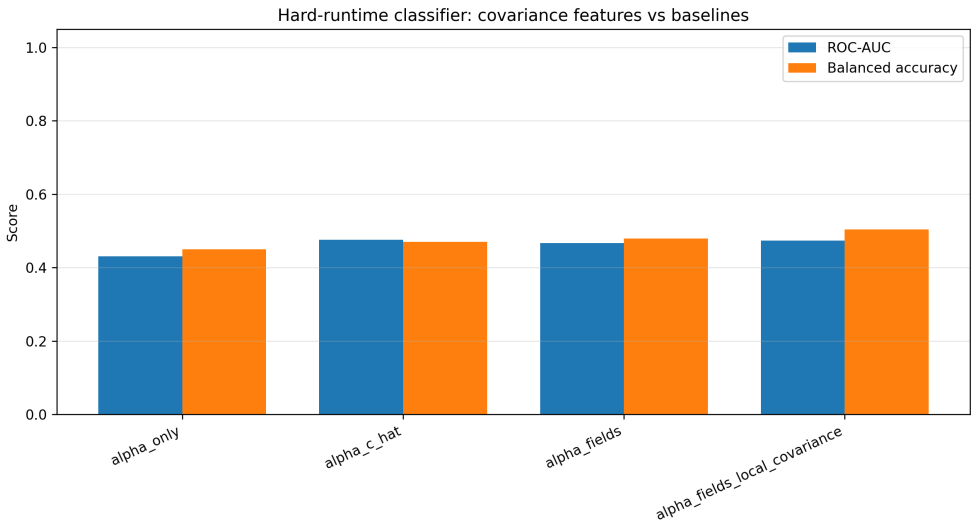

Figure 4: Hard-runtime classifier comparison. The target is top-20% runtime within each α group, not SAT vs UNSAT. Local covariance features yield only a weak improvement in balanced accuracy.

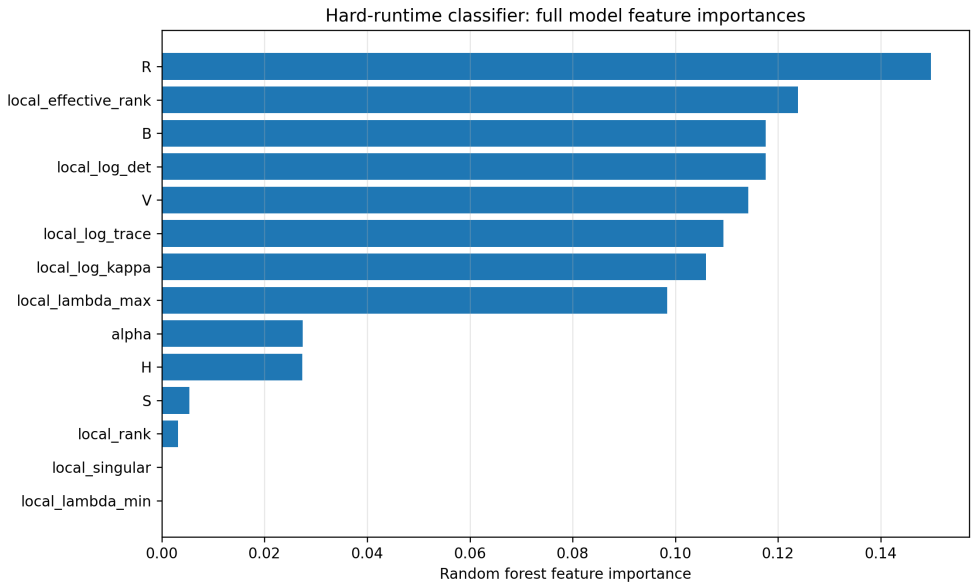

Figure 5: Feature importances in the full classifier. Local covariance observables such as effective rank and log determinant carry some signal, but the present dataset is not sufficient for a strong standalone predictor.

5 Interpretation

The experiments show three things.

First, the benchmark successfully reproduces the expected 3-SAT runtime ridge: runtime peaks near $\alpha \approx 4.26$–4.4 and grows strongly with n.

Second, the MAAT field V acts as a robust within-size structural marker of the ridge. This supports the earlier SAT-atlas observation that connectivity organises the runtime landscape.

Third, covariance geometry is meaningful but subtle. The condition number $\kappa(C)$ separates some ensemble regimes, but it does not behave as a universal scalar hardness law across sizes. The present data instead suggest that covariance geometry should be treated as a higher-dimensional diagnostic containing rank, volume, conditioning, and size-scaling effects.

6 Limitations

The present benchmark has several limitations:

- sample size is modest: 60 formulas per (n, α) cell;
- only one seed schedule is used in the reported scaling run;
- solver runtime is measured by wall-clock time and is therefore noisy;
- formulas are random 3-SAT only; structured industrial SAT instances are not tested;
- the MAAT field definitions are operational proxies rather than unique complexity-theoretic invariants;
- the covariance matrices are often rank-deficient because some fields are nearly frozen in the generated ensembles.

7 Reproducibility

The code, generated data, and figure outputs used for this note are available in the public repository:

https://github.com/Chris4081/sat-hardness-atlas

After cloning the repository, install the required Python packages listed in `requirements.txt`. The numerical pipeline can then be reproduced from the repository root.

Run the local classifier:

```
git clone https://github.com/Chris4081/sat-hardness-atlas
cd sat-hardness-atlas
python3 analysis/sat_hardness_covariance_classifier.py
```

Run the scaling benchmark:

```
python3 analysis/sat_hardness_scaling_test.py --samples 60
```

The main outputs are:

- `results/covariance_classifier/`
- `results/scaling_test/`

The corresponding README files document the generated plots, CSV tables, and JSON summaries. The repository is intended as the authoritative reproducibility bundle for this experiment.

8 Conclusion

This note provides a first MAAT scaling benchmark for structural hardness in random 3-SAT. The strongest positive result is the reproduction of a size-dependent runtime ridge near the expected transition region, together with the observation that connectivity V tracks the ridge within fixed problem size.

The strongest cautionary result is that raw covariance conditioning $\kappa(C)$ is not a universal scalar hardness measure in the present data. Covariance geometry remains important, but it must be treated as an ensemble-level object requiring size normalisation and higher-dimensional analysis.

In short:

> *SAT hardness appears structurally organised, but not reducible to a single covariance condition number.*

References

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[3] D. Mitchell, B. Selman, and H. Levesque, "Hard and Easy Distributions of SAT Problems," in *Proceedings of the Tenth National Conference on Artificial Intelligence*, 459–465 (1992).

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[5] N. Eén and N. Sörensson, "An Extensible SAT-solver," in *Theory and Applications of Satisfiability Testing*, LNCS **2919**, 502–518 (2003).

[6] C. Krieg, "Canonical Definition of the Critical Coherence Index," preprint (2026).

[7] C. Krieg, "Response-Based Derivation of MAAT Structural Weights: From Covariance Geometry to Selection Pressure," preprint (2026).